

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1033

Orinda 100 kV
March 30, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1033. The Interconnection Customer proposes a 241.5 MW solar photovoltaic facility interconnecting at the Orinda 100 kV substation in the Indiana area. The request was received on July 25, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 324 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$142,786,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Roseau County - Orinda 100 kV: 255.3% loading (111 MVA) following the outage of the Harper - Monte Del Estado 6303 100 kV; DFAX = 60.3%.

Lake Marion - Pt A 100 kV: 197.1% loading (172 MVA) following the outage of the Dekalb Substation - Richland 500 kV; DFAX = 10.5%.

For the loss of Roseau County - Orinda 100 kV, the Orinda - Grabill Substation 100 kV L loads to 191.8% of its 118 MVA rating. Project DFAX: 100.0%.

Barret - Point A 100 kV: 188.2% loading (191 MVA) following the outage of the Richland - Dekalb 500 kV line; DFAX = 10.5%.

Grabill Substation - John S Rainey 100 kV: 163.0% loading (127 MVA) following the outage of the Roseau County - Orinda 100 kV; DFAX = 100.0%.

Roseau County - Barret 100 kV: 162.9% loading (246 MVA) following the outage of the Harper - Monte Del Estado 6303 100 kV line; DFAX = 22.6%.

For the loss of Monte Del Estado 6303 - Harper 100 kV, the Monte Del Estado 6303 - Barret 100 kV loads to 152.9% of its 123 MVA rating. Project DFAX: 11.2%.

The Monroeville - Woodburn 100 kV is overloaded to 150.5% of its 230 MVA emergency rating for the loss of the Richland - Dekalb 500 kV. The project contributes 8.4% DFAX (20.4 MW) to this overload.

For the loss of R.P.Mone - Monroeville 100 kV, the Don - Hato Rey Gis 1403-1404 100 kV line loads to 149.4% of its 87 MVA rating. Project DFAX: 5.1%.

The Hesperia - Chub Lake 100 kV L is overloaded to 148.1% of its 131 MVA emergency rating for the loss of the Dekalb - Richland 500 kV line. The project contributes 5.5% DFAX (13.4 MW) to this overload.

For the loss of Richland - Dekalb 500 kV, the Sage - Woodburn 100 kV Line loads to 146.8% of its 244 MVA rating. Project DFAX: 8.4%.

For the loss of Richland - Dekalb 500 kV, the Monte Del Estado 6303 Substation - Sage 100 kV loads to 142.5% of its 264 MVA rating. Project DFAX: 8.4%.

Covanta Hennepin Energy - Chub Lake 100 kV L: 141.7% loading (125 MVA) following the outage of the Richland - Dekalb 500 kV Line; DFAX = 5.5%.

R.P.Mone - Monroeville 100 kV Line: 132.2% loading (104 MVA) following the outage of the Monroeville - Don 100 kV Line; DFAX = 5.3%.

For the loss of Richland - Dekalb Substation 500 kV L, the Segs 5 - Peavine 100 kV Line loads to 131.3% of its 203 MVA rating. Project DFAX: 10.2%.

The John S Rainey - Trier 100 kV L is overloaded to 131.0% of its 138 MVA emergency rating for the loss of the Roseau County - Orinda 100 kV. The project contributes 100.0% DFAX (241.5 MW) to this overload.

For the loss of R.P.Mone - Monroeville 100 kV, the Don - Monroeville 100 kV Line loads to 127.7% of its 113 MVA rating. Project DFAX: 5.1%.

Lakesie 2 - Coanta Hennepin Energy 100 kV L: 124.9% loading (131 MVA) following the outage of the Richland - Dekalb 500 kV; DFAX = 5.5%.

For the loss of Richland - Dekalb 500 kV, the Bark River - San Andres 100 kV L loads to 122.1% of its 164 MVA rating. Project DFAX: 5.9%.

San Andres - Lake Marion 100 kV: 122.1% loading (164 MVA) following the outage of the Dekalb - Richland 500 kV; DFAX = 5.9%.

For the loss of Segs 5 - Hillcrest 100 kV line, the Allen Substation - Wayne Trace 100 kV loads to 116.1% of its 390 MVA rating. Project DFAX: 6.0%.

The Muskogee 230-100 kV TX Bank 1 is overloaded to 113.4% of its 215 MVA emergency rating for the loss of the Richland - Dekalb 500 kV. The project contributes 5.1% DFAX (12.3 MW) to this overload.

The Muskogee 230/100kV Transformer Bank 2 is overloaded to 113.4% of its 215 MVA emergency rating for the loss of the Ricland - Dekalb 500 kV. The project contributes 5.1% DFAX (12.3 MW) to this overload.

The Muskogee 230/100kV Transformer Bank 3 is overloaded to 113.4% of its 215 MVA emergency rating for the loss of the Richland - Dekalb 500 kV L. The project contributes 5.1% DFAX (12.3 MW) to this overload.

The R.P.Mone - Madelia 100 kV L is overloaded to 110.6% of its 117 MVA emergency rating for the loss of the Don - Monroeville 100 kV L. The project contributes 5.3% DFAX (12.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roseau County - Orinda 100 kV	Rebuild line with higher-capacity conductor	\$2,724,000	89% (\$2,421,000)
Lake Marion - Pt A 100 kV	Rebuild line with higher-capacity conductor	\$5,612,000	67% (\$3,741,000)
Grabill Substation - Orinda 100 kV	Rebuild line with higher-capacity conductor	\$11,720,000	100% (\$11,720,000)
Barrt - Point A 100 kV	Rebuild line with higher-capacity conductor	\$10,778,000	67% (\$7,184,000)
Grabill Substation - John S Rainey 100 kV	Rebuild line with higher-capacity conductor	\$3,437,000	100% (\$3,437,000)
Barret - Roseau County 100 kV	Rebuild line with higher-capacity conductor	\$5,154,000	59% (\$3,036,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monte Del Estado 6303 - Barret 100 kV	Rebuild line with higher-capacity conductor	\$8,831,000	100% (\$8,831,000)
Woodburn Substation - Monroeville 100 kV	Rebuild line with higher-capacity conductor	\$5,927,000	58% (\$3,420,000)
Hato Rey Gis 1403-1404 - Don 100 kV	Rebuild line with higher-capacity conductor	\$1,672,000	100% (\$1,672,000)
Hespria - Chub Lake 100 kV	Rebuild line with higher-capacity conductor	\$6,059,000	100% (\$6,059,000)
Sage - Woodburn 100 kV	Rebuild line with higher-capacity conductor	\$3,739,000	58% (\$2,157,000)
Sage - Monte Del Estado 6303 100 kV L	Rebuild line with higher-capacity conductor	\$2,918,000	58% (\$1,684,000)
Covanta Hennepin Energy - Chub Lake 100 kV L	Rebuild line with higher-capacity conductor	\$7,350,000	100% (\$7,350,000)
R.P.Mone - Monroeville 100 kV Line	Rebuild line with higher-capacity conductor	\$8,608,000	100% (\$8,608,000)
Segs 5 - Peavine 100 kV	Rebuild line with higher-capacity conductor	\$1,807,000	54% (\$975,000)
John S Rainey - Trier 100 kV L	Rebuild line with higher-capacity conductor	\$3,788,000	100% (\$3,788,000)
Don - Monroeville 100 kV	Rebuild line with higher-capacity conductor	\$13,423,000	100% (\$13,423,000)
Lakeside 2 - Covanta Hennepin Energy 100 kV line	Rebuild line with higher-capacity conductor	\$4,765,000	100% (\$4,765,000)
Bark River - San Andres 100 kV L	Rebuild line with higher-capacity conductor	\$2,955,000	100% (\$2,955,000)
San Andres - Lake Marion 100 kV	Rebuild line with higher-capacity conductor	\$2,961,000	100% (\$2,961,000)
Allen Substation - Wayne Trace 100 kV	Rebuild line with higher-capacity conductor	\$4,196,000	8% (\$327,000)
Muskogee 230-100 kV TX T1	Replace transformer with higher-rated unit	\$11,000,000	100% (\$11,000,000)
Muskogee 230/100kV Transformer Bank 2	Replace transformer with higher-rated unit	\$11,150,000	100% (\$11,150,000)
Muskogee 230/100kV Transformer Bank 3	Replace transformer with higher-rated unit	\$11,246,000	100% (\$11,246,000)
R.P.one - Madelia 100 kV	Rebuild line with higher-capacity conductor	\$8,876,000	100% (\$8,876,000)
	Total allocated to project		\$142,786,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.